



Proton NMR Spectroscopy

Video Series 1

Video 1: Introduction and Theory



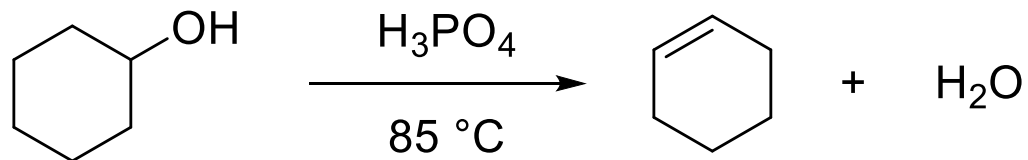


Video 1 Goals

How is ^1H NMR spectroscopy used?

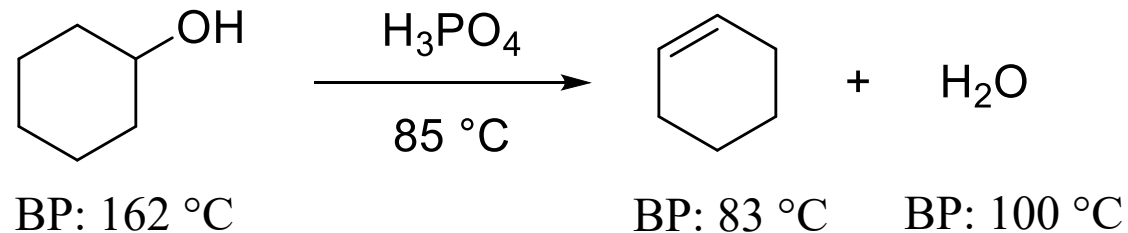
How does ^1H NMR spectroscopy work?

Reaction Analysis and NMR Spectroscopy

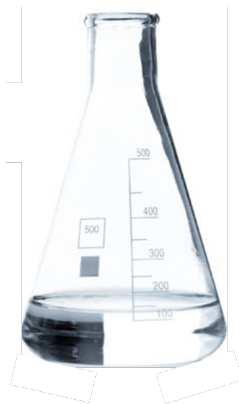


- ☐ Was any desired product formed?
- ☐ Was the starting material completely consumed?
- ☐ Are any undesired side products or impurities present?
- ☐ What is the percent composition of a mixture?

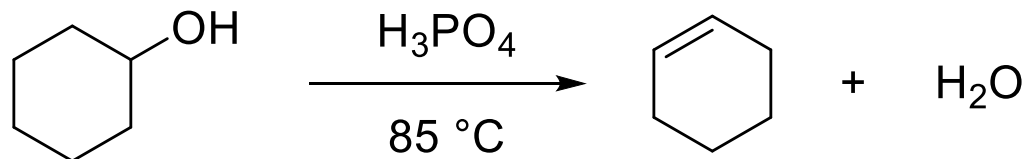
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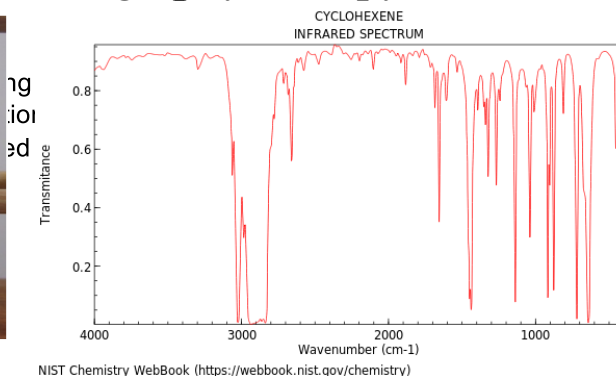
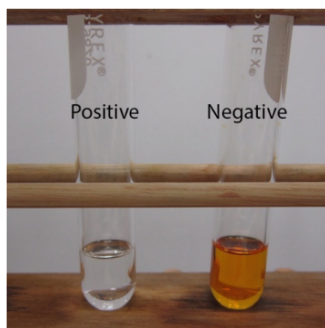


Reaction Analysis and NMR Spectroscopy



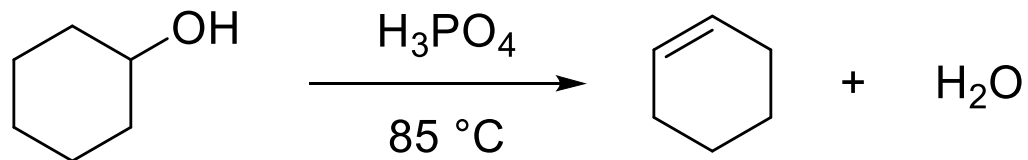
Qualitative Analysis

Thin Layer Chromatography



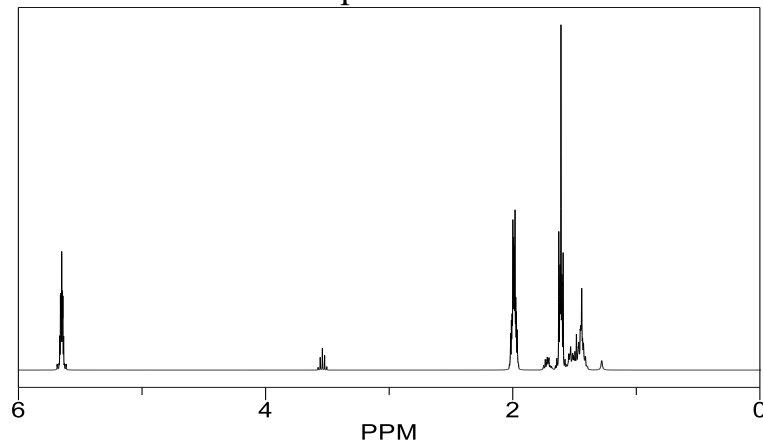
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Reaction Analysis and NMR Spectroscopy



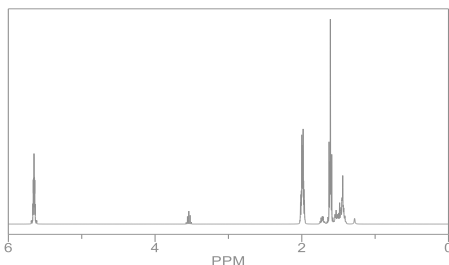
Quantitative Analysis

Simulated ^1H NMR Spectrum



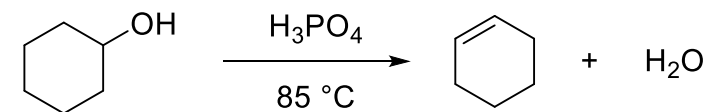
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Reaction Analysis and NMR Spectroscopy



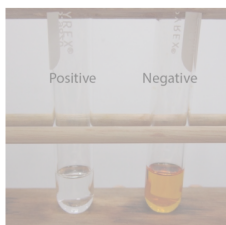
NMR Spectroscopy

- Structure Determination
- Quantitative Analysis

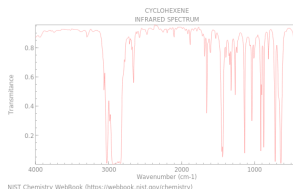


- ☒ TLC Data
- ☒ IR Spectroscopy Data
- ☒ NMR Spectroscopy Data

Chemical Test



IR spectroscopy



TLC



Qualitative Techniques

Chemical tests and IR spectroscopy

- Functional group identification

Thin Layer Chromatography (TLC)

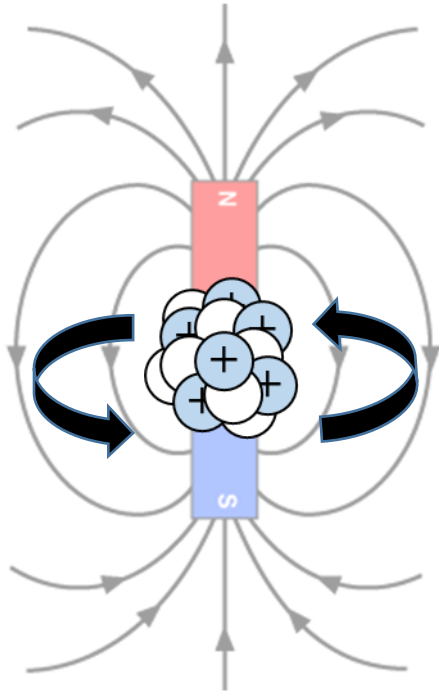
- Identify if undesired impurities are present (no structure information)

Did the reaction work?



How does NMR spectroscopy work?

How does NMR spectroscopy work?



Nuclear spin:

- *Odd* number of protons and *Even* number of neutrons
Even number of protons and *Odd* number of neutrons ($I = 1/2, 3/2, 5/2, \dots$)
e.g. ^1H , ^{13}C , ^{15}N

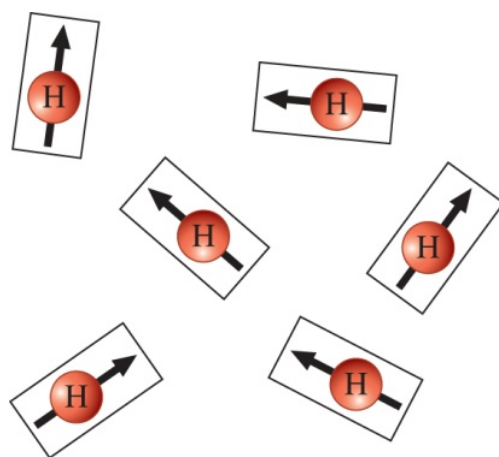
- *Odd* number of protons and *Odd* number of neutrons ($I = 1, 2, 3, \dots$)
e.g. ^2H (Deuterium, D)

NMR Active Nuclei

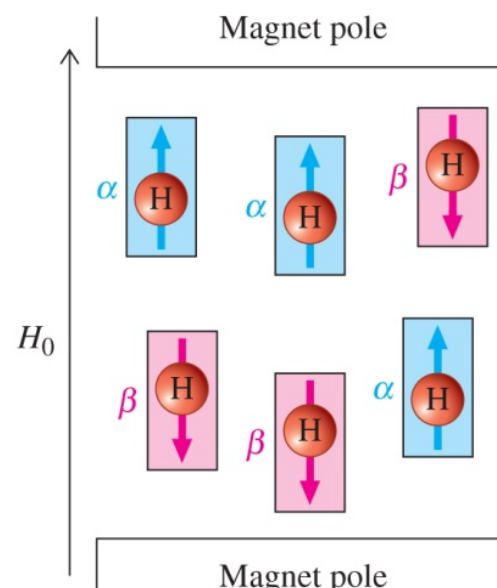
- *Even* number of protons and *Even* number of neutrons ($I = 0$)
e.g. ^{12}C , ^{16}O

NMR Inactive Nuclei

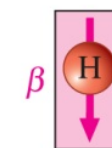
How does NMR spectroscopy work?



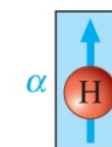
A
No external magnetic
field: random orientation



B
In an external magnetic
field: alignment with
(α) or against (β) H_0

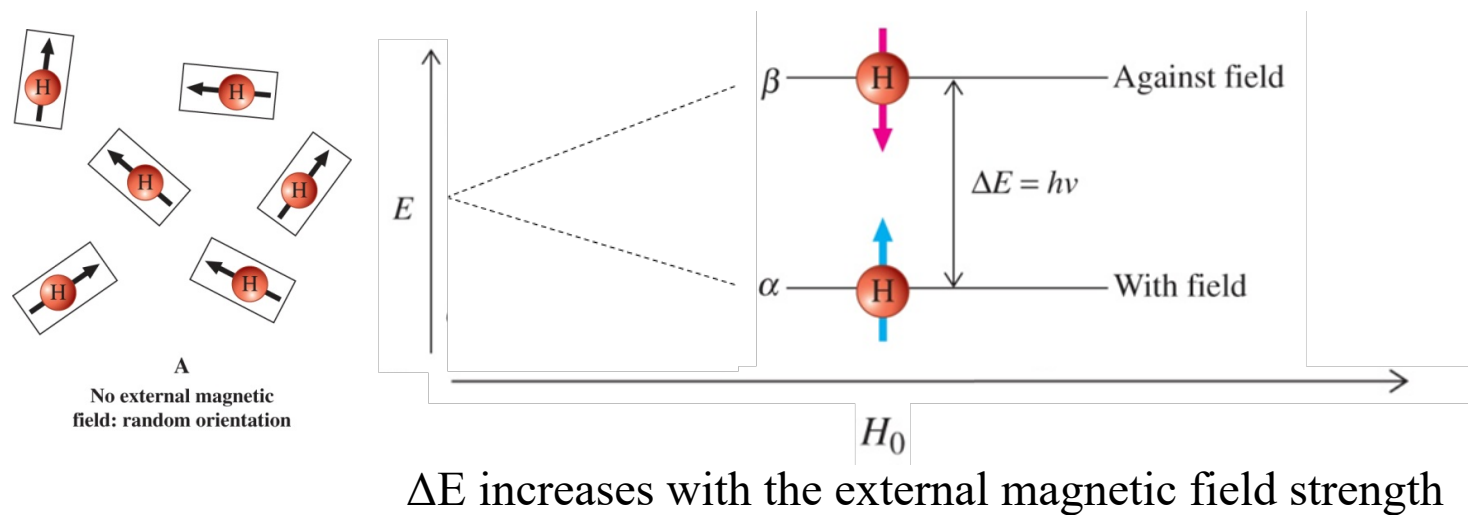


Higher energy
~ 49.9995 %

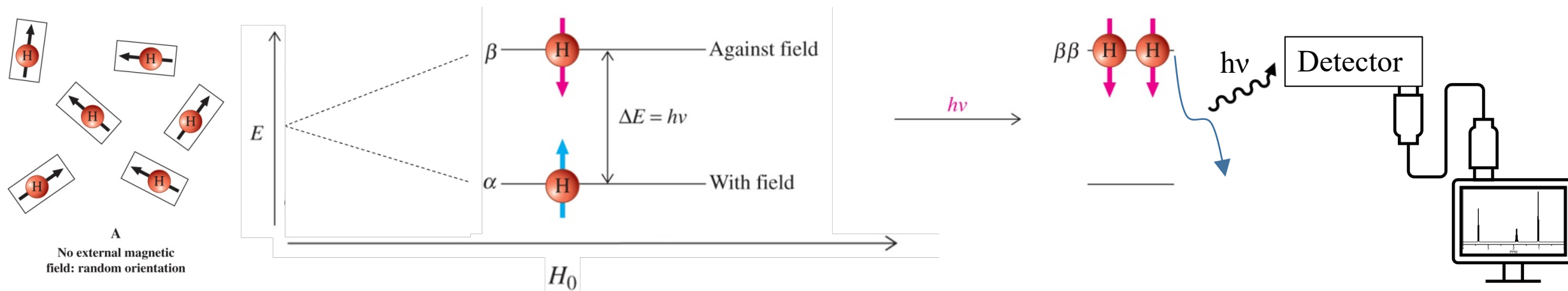


Lower energy
~ 50.0005 %

How does NMR spectroscopy work?

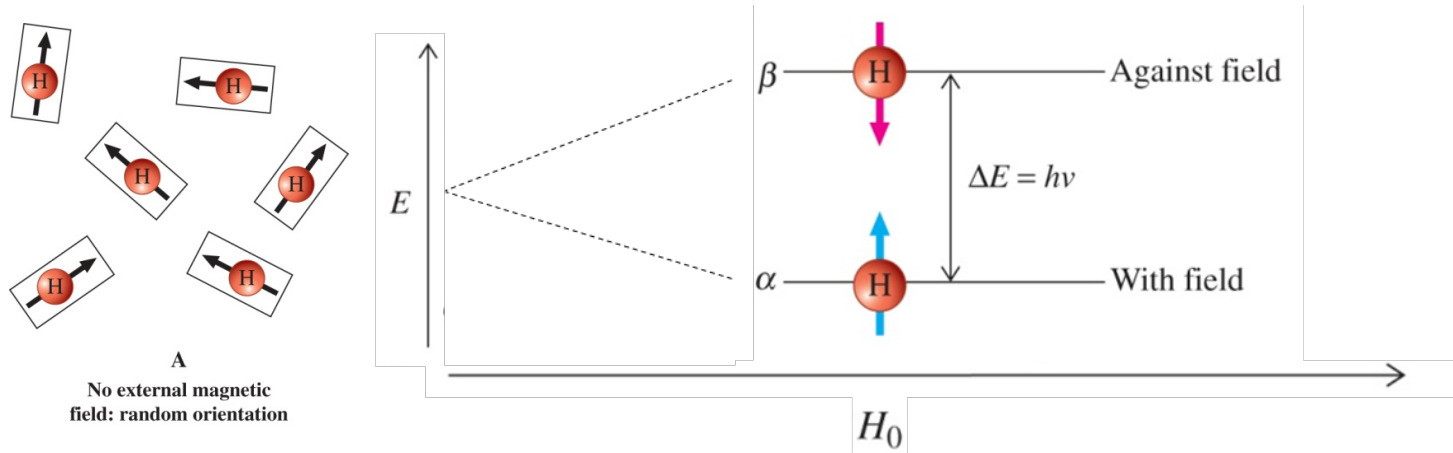


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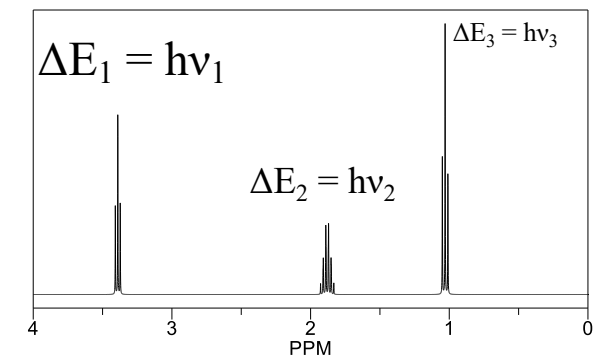
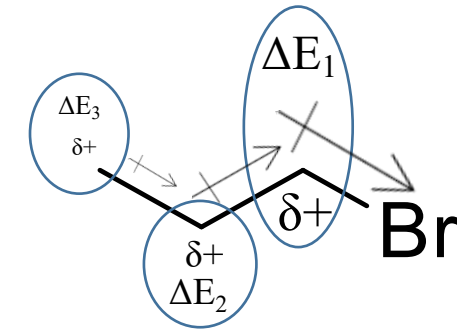


ΔE increases with the external magnetic field strength
 Electron density shields nuclei from the external magnetic field
 ΔE depends on the chemical environment of the nuclei

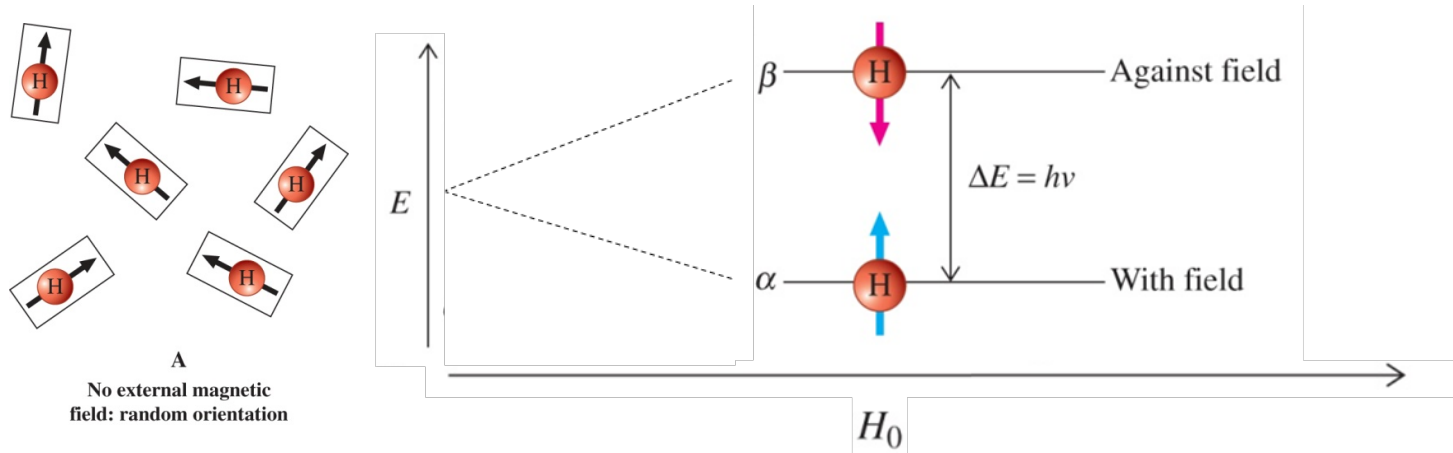
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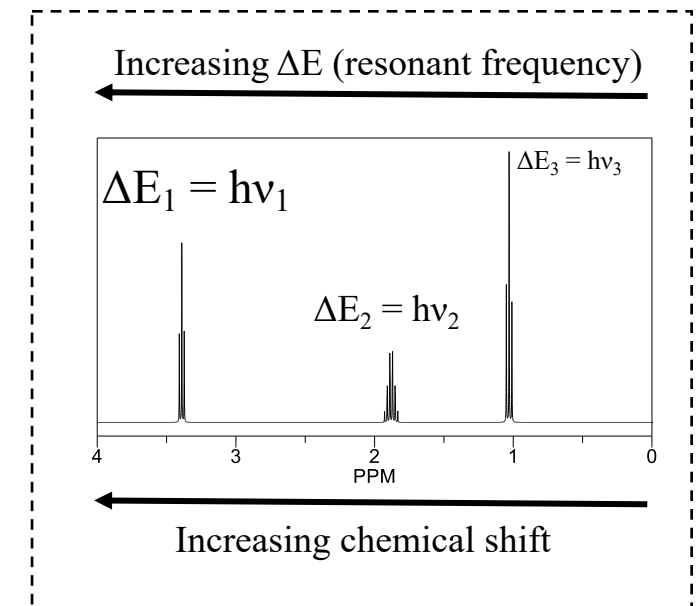
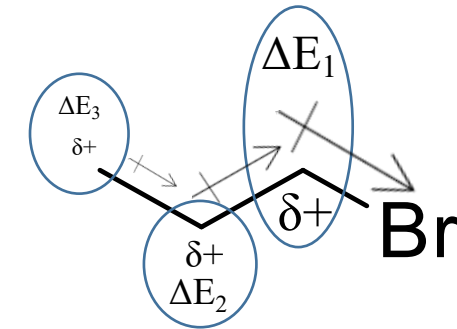
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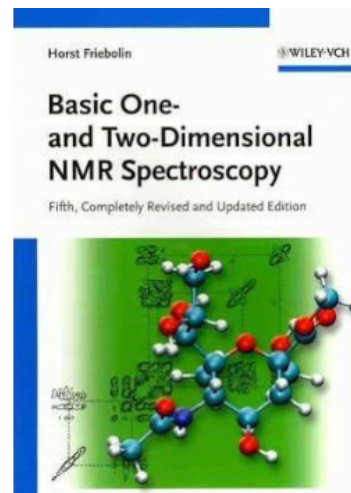
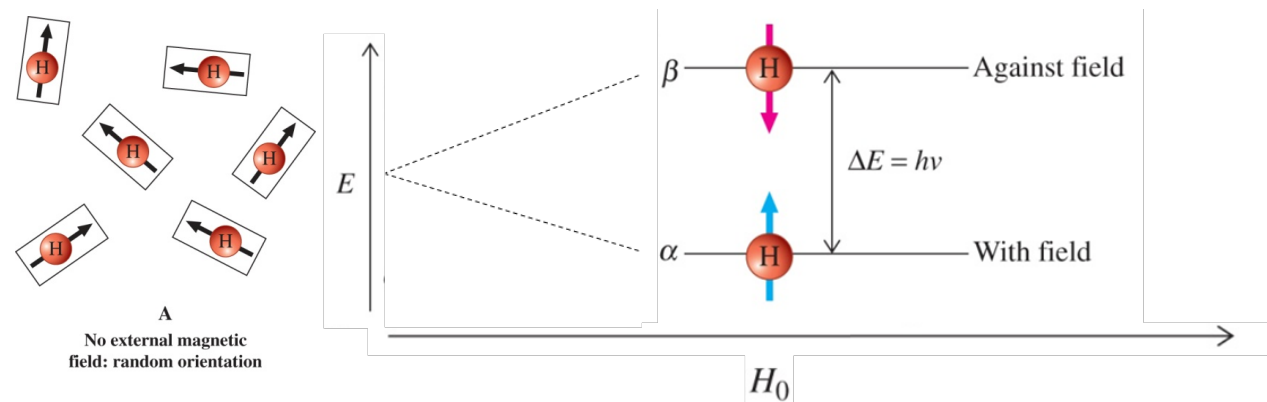
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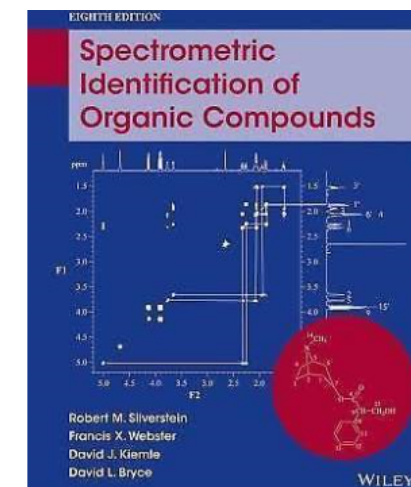
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Further Reading



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